CV

Danny Fainozzi
Current position(s): Junior researcher / Postdoc
Academic age: 3 year(s) 8 month(s)

Education

Degree	Organisation	Duration
PhD / Dr.: Physics Dr. Claudio Masciovecchio	University of Trieste, IT Physics	11.2017 - 10.2020 3 year(s)
Master: Condensed Matter Physics Prof. Fulvio Parmigiani	University of Trieste, IT Physics	10.2015 - 07.2017 1 year(s) 10 month(s)
Bachelor: Physics Prof. Fulvio Parmigiani	University of Trieste, IT Physics	10.2012 - 07.2015 2 year(s) 10 month(s)

Employment

Role	Organisation	Duration
Junior researcher / Postdoc Prof. Andrea Cannizzo	Universität Bern - BE, CH Exakte Wissenschaften	03.2024 - 02.2026 2 year(s)
Junior researcher / Postdoc Dr. Filippo Bencivenga	Elettra-Sincrotrone Trieste S.C.p.A., IT FERMI Free Electron Laser	04.2021 - 03.2024 3 year(s)

Major achievements

Achievement 1

I have over seven years of experience using femtosecond extreme ultraviolet (EUV) and X-ray sources at cutting-edge large-scale research facilities, with a particular emphasis on developing innovative investigation methodologies. Additionally, I have built strong expertise in working with ultrafast optical tabletop setups.

During my PhD and Postdoc, I extensively investigated ultrafast Four-Wave Mixing (4WM) techniques, with a particular focus on the degenerate case of Transient Grating (TG). In a TG experiment, two time-coincident pulses of wavelength λ (the pumps) intersect on the sample at an angle of 2θ, creating a spatial modulation of light intensity with a period $\Lambda = \lambda/(2 \sin \theta)$. This patterned excitation serves as a transient diffraction grating for a third, variably delayed pulse (the probe), producing a fourth, diffracted pulse, the signal. A major challenge in fully characterizing thermoelastic responses at the nanoscale, critical for advancements in faster, more efficient, and more compact nanoelectronic devices, as well as nano-enhanced photovoltaics and thermoelectric energy conversion, was the lack of experimental techniques capable of probing this scale. Recently, Free-Electron Laser (FEL) sources have extended the TG approach into the EUV, enabling contactless excitation and probing of nanoscale thermoelasticity [1].

EUV TG experiments were conducted at the TIMER endstation of the FERMI FEL, where I served as experiments proposer, setup preparator, performer, and data analyst. I often provided custom analysis codes, which were shared with collaborators and students. These efforts led to the publication of over 10 articles, open-sourced whenever possible, with opportunities to visualize the collected data and codes. Among these works, I used EUV TG to demonstrate the ability to obtain Brillouin Backscattering signals [2], achieving a TG period of 6 nm on the sample, the smallest pitch recorded to date. I also successfully tracked heat propagation in a SiO2 sample by probing its thermoelastic response [3], and explored spin wave dynamics, advancing fundamental physics and paving the way for magnonic applications [4]. The latter work was supported by optical Faraday rotation data obtained during a two-month internship at MIT, in Keith Nelson's group.

[1] journal-article. Foglia, L., Mincigrucci, R., Maznev, A. A., Baldi, G., Capotondi, F., Caporaletti, F., Comin, R., De Angelis, D., Duncan, R.A., Fainozzi, D., Kurdi, G., Li, J., Martinelli, A., Masciovecchio, C., Monaco, G., Milloch, A., Nelson, K.A., Occhialini, C.A., Pancaldi, M., Pedersoli, E., Pelli-Cresi, J.S., Simoncig, A., Travasso, F., Wehinger, B., Zanatta, M., & Bencivenga, F. (2023). Extreme ultraviolet transient gratings: A tool for nanoscale photoacoustics. Photoacoustics, 29, 100453. DOI. Open Access.
[2] journal-article. Fainozzi, D., Foglia, L., Khatu, N. N., Masciovecchio, C., Mincigrucci, R., Paltanin, E., & Bencivenga, F. (2024). Stimulated Brillouin Scattering in the Time Domain at 1 nm-1 Wave Vector. Physical Review Letters, 132(3), 033802. DOI. Open Access.
[3] journal-article. Fainozzi, D., Caporaletti, F., Capotondi, F., De Angelis, D., Duncan, R. A., Foglia, L., Martinelli , A., Mincigrucci, R., Nelson, K.A., Pedersoli, E., Zanatta, M., Maznev, A.A., Monaco, G., Bencivenga, F., & Baldi, G. (2024). Temperature effects on the nanoscale thermoelastic response of a SiO2 membrane. APL Materials, 12(5). DOI. Open Access.
[4] journal-article. Miedaner, P.R., Berndt, N., Deschamps, J., Fainozzi, D., Bencivenga, F., Foglia, L., Mincigrucci, R., Paltanin, E., Urazhdin, S., Khatu, N., Brioschi, M., Carrara, P., Bonetti, S., Engel, D.Y., Schick, D., Wittrock, S., Ksenzov, D., Gutt, C., Cucini, R., Comin, R., Nelson, K.A., & Mazenv, A.A. (2024). Excitation and detection of coherent nanoscale spin waves via extreme ultraviolet transient gratings. DOI. Open Access.

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Achievement 2

I made significant contributions to pioneering the hard X-rays Transient Grating (XTG) technique through an X-ray pump/optical probe proof-of-principle experiment [1], supported by optical TG data [2]. In this project, I was responsible for experiment preparation, execution, and data analysis, demonstrating the ability to probe bulk material properties and laying the groundwork for ultrafast coherent 4WM techniques using X-ray probes. The XTG technique has been selected for further development [3] under a Long Term Proposal (LTP) at the European XFEL and will soon become a standard approach at the FXE beamline.

The X-ray pump/X-ray probe (All X-Ray Transient Grating, AXTG) technique was theoretically developed [4] and subsequently demonstrated at SACLA [5], where I took on the roles of experiment preparator, performer, and data analyst. This milestone served as pivoting point in the creation of the COST Action CA22148, an international network for Non-linear EUV to hard X-ray techniques. I coordinated two experimental campaigns at SACLA and LCLS FEL facilities to advance AXTG development. These pioneering efforts are crucial for the materials science community, enabling the measurement of transport properties at nano-ultrafast scales, revealing e.g., the mechanisms underlying novel energy-efficient functional nanomaterials, with applications in nano-thermal devices, ultrafast magnetic switches, organic nanosystems for light harvesting, and ultrafast non-volatile memory storage.

[1] journal-article. Rouxel, J. R., Fainozzi, D., Mankowsky, R., Rösner, B., Seniutinas, G., Mincigrucci, R., Catalini, S., Foglia, L., Cucini, R., Döring, F., Kubec, A., Koch, F., Bencivenga, F., Haddad, A. A., Gessini, A., Maznev, A. A., Cirelli, C., Gerber, S., Pedrini, B., … Svetina, C. (2021). Hard X-ray transient grating spectroscopy on bismuth germanate. Nature Photonics, 15(7), 499–503. DOI. Open Access.
[2] preprint. Fainozzi, D., Catalini, S., Torre, R., Masciovecchio, C., & Svetina, C. (2023). BGO relaxation dynamics probed with heterodyne detected optical transient gratings. arXiv preprint arXiv:2309.02308. DOI. Open Access.
[3] journal-article. Fainozzi, D., Cucini, R., Vila-Comamala, J., Lima, F., Ardana-Lamas, F., Bañares, L., Bargheer, M., Bencivenga, F.,, Berndt, N., Beye, M., Biednov, M., Bonetti, S., Brioschi, M., Cannizzo, A., Carrara, P., Chergui, M., David, C., Engel, R.Y., Ferrari, E., ... & Svetina, C. (2024). Protocols for X-ray transient grating pump/optical probe experiments at X-ray Free Electron Lasers. Journal of Physics B: Atomic, Molecular and Optical Physics. DOI. Open Access.
[4] journal-article. Foglia, L., Mincigrucci, R., Döring, F., Fainozzi, D., Mamyrbayev, T., Masciovecchio, C., & Bencivenga, F. (2024). A new framework for soft x-ray transient gratings. Journal of Physics B: Atomic, Molecular and Optical Physics. DOI. Open Access.
[5] journal-article. Ferrari, E., Ueda, H., Fainozzi, D., Osaka, T., Bencivenga, F., Burian, M., Carrara, P., Vila-Comamala, J., Cucini, R., David, C., Gessini, A., Gerber, S., Goloborodko, A., Hrabec, A., Inoue, I., Inubushi, Y., Leroy, L.D., Roesner, B., Rouxel, J.R., ... & Svetina, C. (2024). All hard X-ray transient grating spectroscopy. Submitted to Nature (2024). DOI. Open Access.

Achievement 3

I demonstrated the capability of performing EUV 3D stereoscopic imaging reconstruction experiments, achieving sub-µm spatial resolution with single-shot capability [1]. In this proof-of-concept experiment, I

served as both the experimenter and data analyst. This advancement paves the way for imaging nanostructures in radiation-sensitive samples, such as molecules, bacteria, and viruses. It also offers an alternative to using multiple identical samples, which inherently reconstruct averaged features.

[1] journal-article. Fainozzi, D., Ippoliti, M., Bille, F., De Angelis, D., Foglia, L., Masciovecchio, C., Mincigrucci, R., Pancaldi, M., Pedersoli, E., Günther, C.M., Pfau, B., Schneider, M., Schmising, C.V.K., Eisebitt, S., Kourousias, G., Bencivenga, F., & Capotondi, F. (2023). Three-dimensional coherent diffraction snapshot imaging using extreme-ultraviolet radiation from a free electron laser. Optica, 10(8), 1053-1058. . DOI. Open Access.

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